DSCPP-III-2026T095

PROJECT AND TEAM INFORMATION

Project Title

Virtual Courtroom Case & Priority System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

The court system has cases that are still waiting to be handled. In court, there are many cases pending that's why it is difficult to manage and decide the hearing dates. Not every case is equally urgent ,some cases need quick attention, like Bail requests, cases involving older people, and serious crime cases. If the court handles cases on a first-come, first-served basis, then urgent and important cases will be delayed.
Our project name is Virtual Courtroom Case Management & Priority System. In our project, all pending cases will be well organized. It will check the urgency and importance of the case, give an urgency score to every case, and cases with a higher urgency score will get higher priority This includes details like what the case's, about the evidence involved people who witnessed the event what happened in earlier hearings and the current status of the case.
The main goal of this project is to show how Data Structures and Object-Oriented Programming can work together to solve a real-life problem in court case management. By making the process of prioritizing and recording cases the system can make the virtual courtroom environment more organized, faster and easier to manage.

State of the Art / Current solution (1 pt)

Today many courts use digital case management systems to store case records, schedule hearings, track case status and manage documents.. Real judicial systems are very complex because they involve legal procedures, court rules, judges, lawyers and administrative staff.

Modern digital systems help organise amounts of case information.. Deciding the priority of cases can depend on many different factors. Urgent cases such as bail matters or cases involving individuals may require faster attention.

Our project takes the concept of digital case management and creates a smaller educational simulation. I think this project is exciting. The main focus is, on managing pending cases and automatically identifying which case should receive higher priority.

The system will use an urgency score, priority queue, hash map, linked list and other data structures to demonstrate case management. Unlike court systems our project is designed purely for educational purposes and uses simulated case data.

Project Goals and Milestones (2 pts)

The main goal of our project is to develop a Courtroom Case Management & Priority System that can efficiently manage pending cases and determine their hearing priority.

The user will be able to register types of cases such as Criminal, Civil and Family cases. Important information like case severity, pending days, special conditions, evidence and witnesses will be stored in the system.

Each case will receive an urgency score. Based on this score the system will determine which case should be considered first. A Priority Queue will be used to maintain the cases according to their urgency.

A Hash Map will allow searching of cases using a unique Case ID. A Linked List will maintain the hearing history and proceedings of each case. A Stack can also be used to implement an undo feature for recent actions.

For the part we will use classes such as Case, CriminalCase, CivilCase, FamilyCase and CourtSystem. Concepts such, as encapsulation, inheritance, abstraction and polymorphism will be implemented.

The major milestones include case registration, urgency calculation, priority queue implementation, case searching, hearing-history management, testing and final project development.

Project Approach (3 pts)

We plan to build the project using C++. The user will start by entering information about a case. This information includes the type of case how serious it is, how days it has been waiting and if there are any special things that need attention.

The system will figure out an urgency score for each case. This score could be based on things like how the case has been waiting how serious it is, if an older person is involved and if the case is about getting out of jail.

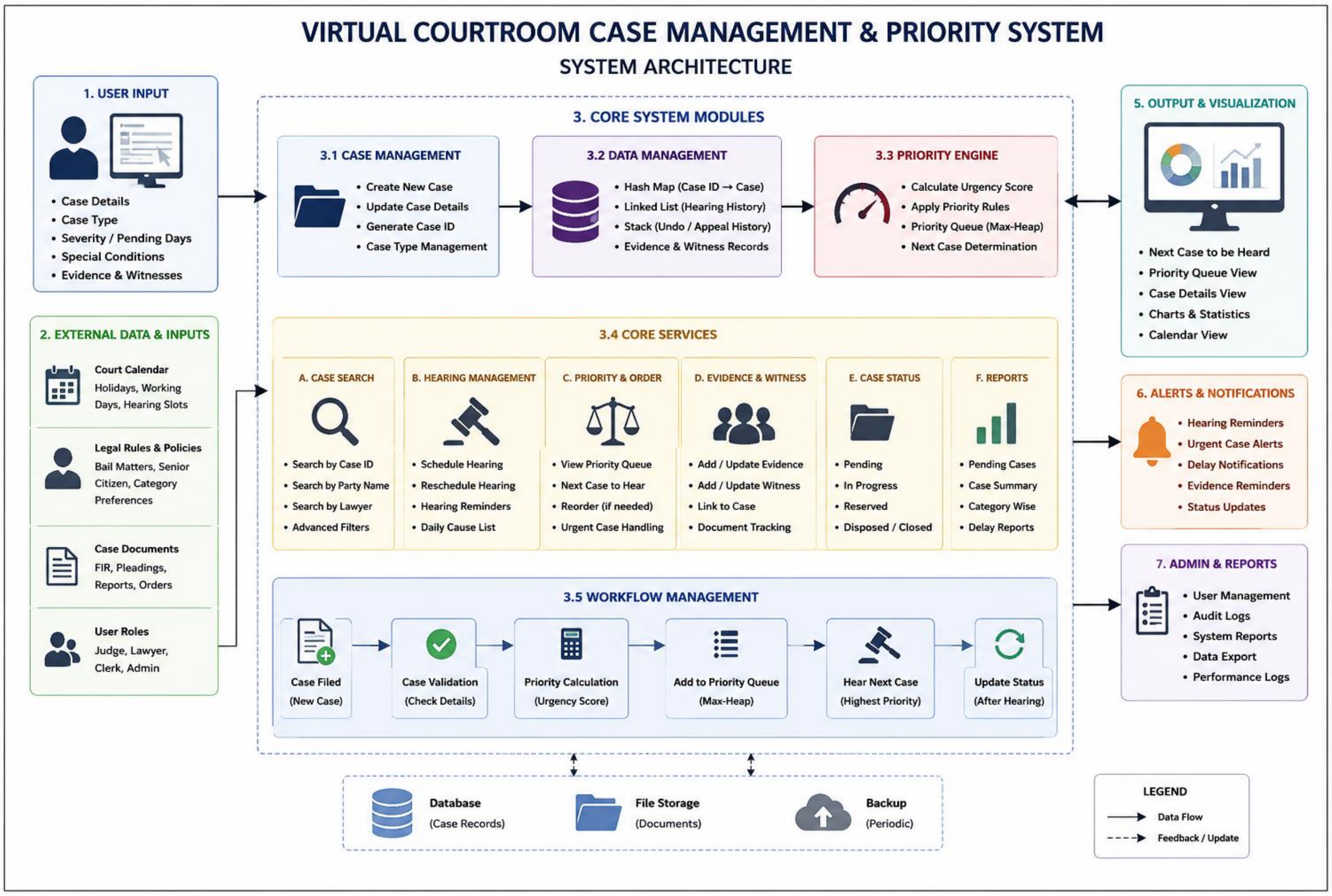
A Priority Queue that uses a Max-Heap will hold all the cases that are waiting. The case with the urgency score will always be at the top. This makes it easy to see which case should be handled next.

A Hash Map will help find cases quickly using their Case ID. A Linked List will keep track of the history of hearings the evidence, the people who gave testimony and what happened during each step of the case. A Stack can be used to keep track of actions and allow users to undo those actions.

For the Object-Oriented Programming part we will create a base class called Case. Different types of cases, like CriminalCase, CivilCase and FamilyCase will inherit from this class. Each type of case can have its way of calculating priority by using polymorphism.

The project will be. Tested with C++ and VS Code. The final system will have a way for people to register cases look them up set their priority make changes and finish them.

System Architecture (High Level Diagram)(2 pts)



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Project Outcome / Deliverables (1 pts)

The main outcome of our project is a functional Virtual Courtroom Case Management & Priority System that keeps pending cases organised and runs smoothly.

The Courtroom Case Management & Priority System lets users register new cases and save key information about them. It also calculates an urgency score so that the Virtual Courtroom Case Management & Priority System knows which case deserves priority.

The Courtroom Case Management & Priority System will have many useful features. These include registering cases searching for a case by its Case ID ordering hearings, by priority, managing evidence and witnesses tracking hearing history and updating case status.

The Courtroom Case Management & Priority System will show how important Data Structures work. It will use Priority Queues, Hash Maps, Linked Lists and Stacks in ways.

Another key goal is to use Programming ideas. Different kinds of cases will be shown with inheritance and polymorphism.

By the end of the project we want to build an educational simulation that shows how Data Structures and OOP can team up to solve a real case-management problem.

Assumptions

We believe that all of the case information that the user gives us is valid and reasonable. The system will utilise case data rather than actual court records.
Hook delay, severity and special categories are some of the factors that we believe could be assigned values to compute the urgency score.
The priority calculation shown in this project is just for educational purposes. It does not represent the real judicial rules and procedures.
This is only a demo model to show how things could work . It is not made to replace real judges, lawyers or real court management systems

References

- GeeksforGeeks :- *Data Structures and Algorithms in C++.*
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